Cancellation zeros and the next recoil terms in the full Barnes amplitude

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Abstract

The two-branch Bessel leading term has infinitely many simple zeros on the positive real parameter plane. We compute the next integrated coefficients of the exact Barnes kernel and show that these auxiliary zeros are lifted at order ν^{-2}. Thus a uniform relative error dividing by the leading sum cannot include this set. Away from cancellation, the second coefficient explains the previously observed K^3/ν^2 scale. The integrated expansion is proved at fixed parameters; a joint growing-K remainder remains open.

1 Object and a fixed-parameter expansion

Use radius one, $u = v = 0$, $\omega = i\alpha$, $K > 0$, real α, $M = \nu^2/(2K)$. The object is the internal chiral graph amplitude after the fixed-mass $\kappa \to \infty$ limit. No finite-G state, physical external insertion or normalized OTOC is identified here. Set $t = z + i(M + \nu)$,

$$b = \tfrac{1}{2} - i(M - \nu), \quad c = \tfrac{1}{2} - i(M + \nu), \quad C = e^{-\pi(M+\nu)} \frac{\Gamma(-i\nu/2)^4 \Gamma(b)}{\Gamma(c)}.$$

The normalized kernel, inherited from the preceding full-limit note, is

$$Q_\nu(t) = e^{-i\pi t} \frac{\Gamma(t - i\nu/2)^4}{\Gamma(-i\nu/2)^4} \frac{\Gamma(b - t)\Gamma(c)}{\Gamma(b)\Gamma(c + t)} \Gamma(\tfrac{1}{2} - t + \alpha/2)\Gamma(\tfrac{1}{2} - t - \alpha/2).$$

Its contour runs upward from $-N - i\infty$ to $-N + i\eta$, then right, with $\eta > 0$, integer $N > |\alpha|/2 + 1/2$. Write $\mathcal{I}_\nu = (2\pi i)^{-1} \int Q_\nu \, \mathrm{d}t = \mathfrak{B}/C$. Define, with $\mathcal{D} = (K/2)\partial_K$ at fixed α,

$$J = -K[e^{i\pi\alpha} K_\alpha(K) + i\pi I_\alpha(K)], \qquad S = K^2 + \alpha^2 - 1,$$
$$T = \tfrac{1}{3}[2S\mathcal{D}J + (K^2 - \alpha^2 + 1)J], \qquad P = \tfrac{1}{6}[9K^4 - (\alpha^2 - 1)^2].$$

Theorem 1. *For fixed K, α, as $\nu \to \infty$,*

$$\mathcal{I}_\nu = e^{i(S/\nu + P/\nu^3)} \left(J + \frac{T}{\nu^2} \right) + O_{K,\alpha}(\nu^{-4}). \tag{1}$$

The error is absolute, including at $J = 0$.

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2 Derivation of the integrated coefficients

Let

$$U(t) = e^{-2\pi i t}(K/2)^{2t}\Gamma(\tfrac{1}{2} - t + \alpha/2)\Gamma(\tfrac{1}{2} - t - \alpha/2).$$

On the logarithmically growing truncation $R = \log \nu$, Gamma-ratio expansions [1] on safe sectors give

$$\log(Q_\nu/U) = \frac{L_1}{\nu} + \frac{L_2}{\nu^2} + \frac{L_3}{\nu^3} + O_{K,\alpha}((1+|t|)^5/\nu^4), \tag{2}$$

where log ratios are continued from zero shift, and

$$\begin{aligned} L_1 &= 4i(t^2 - t), \\ L_2 &= \tfrac{8}{3}B_3(t) + 4K^2 t = \tfrac{8}{3}t^3 - 4t^2 + (\tfrac{4}{3} + 4K^2)t, \\ L_3 &= 4iK^2t^2 - \tfrac{8i}{3}(t^2 - t)^2. \end{aligned}$$

For the four arguments $-i\nu/2$, the Bernoulli terms through B_4 give the displayed cubic and quartic polynomials. The leading b, c powers contribute $(4K^2 - 2iK)t/\nu^2$; their quadratic shifts supply $2iKt/\nu^2 + 4iK^2t^2/\nu^3$. This explicitly cancels the imaginary second-order term.

Exponentiating (2), the coefficient polynomials are

$$A_2 = L_2 + \tfrac{1}{2}L_1^2 = -8t^4 + \tfrac{56}{3}t^3 - 12t^2 + (\tfrac{4}{3} + 4K^2)t, \quad A_3 = L_3 + L_1L_2 + \tfrac{1}{6}L_1^3. \tag{3}$$

Replace t^n under the auxiliary integral by $\mathcal{D}^n$, keeping the displayed coefficient powers of K to the left. Since $\mathcal{D}^2 J = \mathcal{D}J + (S/4)J$, every result reduces to $J, \mathcal{D}J$. For a direct algebraic verification, if $q = S/4$, then

$$\mathcal{D}(FJ + G\mathcal{D}J) = (\mathcal{D}F + qG)J + (F + \mathcal{D}G + G)\mathcal{D}J, \qquad \mathcal{D}q = K^2/4.$$

This gives

$$A_2(\mathcal{D})J = T - \tfrac{1}{2}S^2 J, \qquad A_3(\mathcal{D})J - iSA_2(\mathcal{D})J - \tfrac{i}{3}S^3 J = iPJ. \tag{4}$$

The third term after removal of the first phase is thus proportional to J, while the second contains its derivative.

For completeness, the preceding tail proof supplies, uniformly in large ν at fixed K, α,

$$|Q_\nu(-N - is)| \le C(1+s)^{2N}e^{-3\pi s}, \quad \int_R^\infty |Q_\nu(x + i\eta)|\,\mathrm{d}x \le 6RH_R,$$

with $H_R = \sup_{[R,R+1]} |Q_\nu(x + i\eta)|$ once $R \ge \max(2, 2K, 2\max(\alpha^2/4 - \eta^2, 0))$. At $R = \log\nu$, (2) implies $H_R \le C\exp[-2R\log R + O_K(R)]$. The vertical tail is $o(\nu^{-4})$, the horizontal tail is smaller than every fixed inverse power, and all polynomial moments of $|U|$ are integrable. The truncated exponentiation error is bounded by $C\nu^{-4}(1+|t|)^8|U|$. Integration and (4) prove (1). The constants are not joint-K bounds.

3 The leading sum has simple real cancellation zeros

It suffices to take $\alpha \geq 0$, since the complete J is even in α. For $K > 0$, $I_\alpha(K)$ and $K_\alpha(K)$ are positive [3]. Hence

$$\Re J = -K\cos(\pi\alpha)K_\alpha(K).$$

A zero requires $\cos(\pi\alpha) = 0$. If $\sin(\pi\alpha) = +1$, the imaginary terms add. Therefore all positive-order zeros occur at

$$\alpha_n = 2n + \tfrac{3}{2}, \quad n = 0, 1, \ldots, \qquad K_{\alpha_n}(K_n) = \pi I_{\alpha_n}(K_n). \tag{5}$$

There is exactly one $K_n > 0$ for each such α_n. Indeed the ratio $r(K) = K_\alpha(K)/I_\alpha(K)$ runs from infinity to zero, and the standard Wronskian [2] gives

$$r'(K) = -\frac{1}{KI_\alpha(K)^2} < 0.$$

At the zero,

$$J'(K_n, \alpha_n) = -\frac{i}{I_{\alpha_n}(K_n)}, \qquad T(K_n, \alpha_n) = -\frac{iK_nS_n}{3I_{\alpha_n}(K_n)} \neq 0, \quad S_n = K_n^2 + \alpha_n^2 - 1. \tag{6}$$

Equation (1) then implies, at each fixed zero,

$$\mathcal{I}_\nu(K_n, \alpha_n) = -\frac{iK_nS_n}{3I_{\alpha_n}(K_n)\nu^2}e^{iS_n/\nu} + O_{K_n,\alpha_n}(\nu^{-4}). \tag{7}$$

The exact integral therefore does not retain that auxiliary zero for all sufficiently large ν. This is not a claim locating all zeros of the finite-mass integral or excluding displaced zeros.

The zeros accumulate near the previously found magnitude crossover. Let $F(y) = \sqrt{1+y^2} - y\,\mathrm{arsinh}\,y$, with positive root $y_c \simeq 1.50887956$. Uniform Bessel asymptotics [4] give, near y_c,

$$\log\frac{K_\alpha(K)}{\pi I_\alpha(K)} = -2KF(\alpha/K) + O(K^{-1}).$$

Monotonicity first traps the root there, and then

$$\alpha_n/K_n = y_c + O(\alpha_n^{-2}). \tag{8}$$

These are auxiliary interference zeros, not physical critical points.

α_n	K_n (approximate)	$\alpha_n^2(\alpha_n/K_n - y_c)$
1.5	1.00000000	−0.0199790
3.5	2.32218535	−0.0205628
7.5	4.97178686	−0.0206769
15.5	10.27310967	−0.0207018
31.5	20.87670645	−0.0207076
63.5	42.08435037	−0.0207089

4 Consequences for the joint recoil window

Any relative error divided by J is undefined on (5), and (7) shows that this is not just a harmless shared zero. A prospective uniform result should instead control the complete complex amplitude, for example against the positive branch envelope

$$\mathcal{E}(K,\alpha) = K[K_\alpha(K) + \pi I_\alpha(K)]. \tag{9}$$

A bound relative to $\mathcal{E}$ has not yet been proved for the joint limit. It would preserve absolute control near interference and imply a relative estimate only where $|J|$ is bounded below by a suitable fraction of $\mathcal{E}$.

Away from cancellation, set

$$c_2(K,\alpha) = T(K,\alpha)/J(K,\alpha) - T(K,0)/J(K,0).$$

For $\alpha = Ky$, fixed $y > 0$ separated from y_c, Bessel derivative asymptotics at *fixed order when differentiating* give the coefficient limits

$$\frac{c_2(K,Ky)}{K^3} \longrightarrow \begin{cases} ((1+y^2)^{3/2} - 1)/3, & 0 < y < y_c, \\ -((1+y^2)^{3/2} + 1)/3, & y > y_c. \end{cases} \tag{10}$$

The reference $y = 0$ contributes $+1/3$. Equation (10) is an asymptotic of an explicitly known coefficient, not of the unbounded remainder in (1). It identifies the candidate small scale

$$K^3/\nu^2 = \nu^4/(8M^3).$$

Thus $M \gg \nu^{4/3}$ suppresses this coefficient, while $M \ll \nu^2$ ensures $K \gg 1$. This agrees with the earlier candidate recoil window but does not prove its sufficiency. The first phase K^2/ν can be large in that window, so a joint analysis must retain it rather than truncate it.

At fixed parameters, since $J(K,0) \neq 0$, (1) also yields the absolute normalized-ratio approximation

$$\frac{\mathfrak{B}(K,\alpha)}{\mathfrak{B}(K,0)} = e^{i[\alpha^2/\nu - (\alpha^4 - 2\alpha^2)/(6\nu^3)]} \frac{J(K,\alpha) + T(K,\alpha)/\nu^2}{J(K,0) + T(K,0)/\nu^2} + O_{K,\alpha}(\nu^{-4}). \tag{11}$$

This form retains the derivative term even when the numerator's leading J vanishes.

Reusing the frozen full-integral data at $\nu = 4K^3$ gives:

K	y	first phase only	plus c_2/ν^2	formula (11)
4	2	$3.89\,10^{-3}$	$4.02\,10^{-5}$	$6.35\,10^{-6}$
8	1	$7.43\,10^{-5}$	$7.71\,10^{-8}$	$6.64\,10^{-9}$
8	2	$4.93\,10^{-4}$	$1.27\,10^{-6}$	$1.02\,10^{-7}$
16	2	$6.19\,10^{-5}$	$3.97\,10^{-8}$	$1.60\,10^{-9}$

Entries are relative numerical differences; no coefficient was fitted.

5 A joint positive-tail ingredient at an $O(K)$ cutoff

The earlier recurrence bound began at $A = O(\alpha^2)$. This threshold can be bridged under the same candidate hierarchy, without raising the contour height. Fix $Y \geq 0$, $K \geq 1$, $|\alpha| \leq KY$, and $\eta = 1$. Set

$$D_Y = \max(2, Y^2/2), \quad c_Y = 8 + 2Y, \quad A = \max(2, 2K, 2d), \quad d = \max(\alpha^2/4 - 1, 0).$$

Assume

$$\nu^2 \geq 32(D_Y + 1)K^3, \qquad R \geq c_Y K. \tag{12}$$

The exact recurrence from the preceding note is

$$\mathcal{R}_\nu(t) = \frac{Q_\nu(t+1)}{Q_\nu(t)} = \frac{(t - i\nu/2)^4}{[(t + \frac{1}{2} - i\nu)^2 + M^2][(t + \frac{1}{2})^2 - \alpha^2/4]}.$$

If $R \leq A + 1$, write $a = \nu/2 - 1$, $L = M - \nu + 1$. The earlier denominator bounds, $L \geq \nu^2/(4K)$, and $x^2 - d \geq x^2/2$ give, for $R \leq x \leq A + 1$,

$$\begin{aligned} |\mathcal{R}_\nu(x + i)| &\leq \frac{2(x^2 + a^2)^2}{L^2 x^2} \\ &\leq 2\left(\frac{A+1}{L} + \frac{a^2}{RL}\right)^2 \leq 2\left(\frac{1}{8} + \frac{1}{8}\right)^2 = \frac{1}{8}. \end{aligned}$$

Here $A + 1 \leq (D_Y + 1)K^2$, so the first fraction is at most $4(D_Y + 1)K^3/\nu^2$; the second is at most K/R. Assumption (12) also implies the earlier condition $\nu \geq 8K$. For $x \geq A$, the already proved step bound $|\mathcal{R}_\nu(x + i)| \leq [x/(x + 1)]^{3/2}$ applies.

Thus the geometrically contracting transition reaches the polynomially bounded far tail. If $n_0 = \max(0, \lceil A - R \rceil)$, the first n_0 unit steps cost at most 8^{-n_0}, and the remaining tail is bounded by $6(R + n_0)H_R 8^{-n_0}$. Summing the finite transition gives the convenient universal estimate

$$\int_R^\infty |Q_\nu(x + i)| \, \mathrm{d}x \leq 10(R + 1)H_R, \qquad H_R = \sup_{R \leq x \leq R+1} |Q_\nu(x + i)|. \tag{13}$$

If $R \geq A$, the original far-tail bound immediately implies (13).

This is a joint-K positive-tail lemma: the base cutoff need only be $O_Y(K)$, rather than $O_Y(K^2)$. It does not bound H_R relative to the branch envelope, nor the negative contour with $N = O(K)$. Those two tasks, together with the integrated local remainder after phase retention, remain necessary for a full joint asymptotic theorem. Eighteen finite samples check the new contraction inequality, including masses at the threshold in (12).

6 Direct checks at zeros and next proof requirements

At the first two auxiliary zeros we directly integrate the exact Q_ν, retaining both complex contour segments. For the leading nonzero prediction in (7), we find

α	ν	relative error against (7)'s leading term
1.5	64	$3.97013\,10^{-4}$
1.5	128	$9.91999\,10^{-5}$
1.5	256	$2.47966\,10^{-5}$
3.5	64	$9.38078\,10^{-4}$
3.5	128	$2.32015\,10^{-4}$
3.5	256	$5.78474\,10^{-5}$

The segment cancellation ratio reaches about $1.70\,10^7$. We use 20/32 Gauss nodes per segment and 60/80 digits, with a 100-digit comparison at the largest ν. The latter also changes $(X, T, \eta) = (36, 18, 1)$ to $(44, 24, 0.75)$. Refinements compare the complete complex sum. These are empirical diagnostics, not interval certification.

All 65 checks pass: 3 symbolic operator identities, 4 compact remainder-decrease checks, 18 zero/Wronskian/lifting identities, 6 quadrature refinements, 4 precision/contour comparisons, 4 zero-lifting error decreases, 8 improvements in reused joint samples, and 18 joint-tail samples. Six compact residuals and six reused fixed-parameter second-order residuals are also stored. Code, versions, parameters, complex segments, and input SHA256 hashes are in the accompanying project records.

The next uniform theorem must use an absolute or branch-envelope error and keep the derivative correction through cancellation. A promising route is to split the two analytic contributions before bounding their sum, then prove contour estimates with explicit K, α dependence after retaining the first phase. The positive-tail transition is now controlled by (13); the base size and the negative contour still require estimates with explicit parameter dependence. The joint remainder, transverse variables, physical external energy cycles and normalization, disconnected baseline, finite-G limit, and microscopic dictionary all remain open. This is an author self-reviewed proof draft.

References

[1] NIST DLMF, §5.11, Gamma asymptotics, https://dlmf.nist.gov/5.11.

[2] NIST DLMF, 10.28.2, Bessel Wronskian, https://dlmf.nist.gov/10.28.

[3] NIST DLMF, §10.30, §10.32, and §10.40, positivity and limiting forms, https://dlmf.nist.gov/10.32.

[4] NIST DLMF, §10.41(ii), including derivative expansions, https://dlmf.nist.gov/10.41.